

# Jasmin Di Franco

PhD Researcher in Experimental Soft Matter Physics & Biophysics

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## Research Profile

PhD researcher in experimental soft matter physics and biophysics with a background in chemical engineering. My research examines phase transitions, collective dynamics, and mechanical response in dense biological systems. I combine mammalian and bacterial cell culture with long-term quantitative microscopy, particle tracking, image analysis, and physical modeling. My work is interdisciplinary and connects experiments with theory and simulation through national and international collaborations.

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## Research Experience

### PhD Researcher in Experimental Biophysics

Nov 2022–Dec 2026

*University of Vienna, Faculty of Physics*

Vienna, Austria

Doctoral research in the group of Prof. Roberto Cerbino on jamming and unjamming, structure–dynamics relationships, and collective motion in dense cellular systems.

- Lead a comparative study of epithelial and fibroblast monolayers to test the universality of cellular jamming, combining cell and nuclear morphology, collective motion, relaxation dynamics, static structure, adhesion-focused experiments, and controlled simulations.
- Design and conduct experiments spanning mammalian cell culture, sample preparation, and automated 24–72-hour quantitative live-cell imaging.
- Develop reproducible workflows for segmentation, morphology analysis, particle tracking, particle image velocimetry (PIV), differential dynamic microscopy (DDM), and time-resolved microscopy data.
- Carried out quantitative analysis of collective cellular dynamics to support collaborative studies of vocal-fold cancer mechanobiology and connexin-mediated flocking transitions ([cellpiv](#): cellular dynamics workflow).
- Contributed to validating culture protocols for *Myxococcus xanthus* for an open-access tutorial to promote their reproducible use by the research community.
- Collaborated with national and international partners to integrate experimental findings with physical modeling and computational simulations.

### Visiting Researcher

Nov 2025

*Cellular Biomechanics, University of Bayreuth*

Bayreuth, Germany

Host group: Prof. Elisabetta Ada Cavalcanti-Adam.

- Cultured cell lines with tunable adhesion properties.
- Conducted experiments under controlled adhesion conditions to investigate the effect on cellular dynamics.

### Visiting Researcher

Oct–Nov 2023

*Mechanisms of Tumor Cell Migration Unit, IFOM*

Milan, Italy

Host group: Prof. Giorgio Scita.

- Received training in mammalian cell-culture techniques.
- Performed experiments investigating collective cell dynamics and jamming.

### Master's Thesis in Product Engineering

2022

*University of Naples Federico II*

Naples, Italy

*Mechanical Behavior of Tumor Spheroids: Shear-Induced Surface Erosion.*

- Investigated tumor-spheroid morphology and flow-induced surface erosion under controlled simple shear.

## Education

### PhD in Physics

University of Vienna

Expected 2027  
Vienna, Austria

Supervisor: Prof. Roberto Cerbino, Somexlab 

Thesis: *Phase Transitions and Collective Dynamics in Dense Biological Systems*.

Submission planned for December 2026; defense expected in spring 2027.

### MSc in Product Chemical Engineering

University of Naples Federico II

2020–2022  
Naples, Italy

### BSc in Chemical Engineering

University of Naples Federico II

Completed 2020  
Naples, Italy

## Research Interests

Soft matter; cellular jamming and unjamming; collective cell dynamics; active matter; mechanobiology; quantitative microscopy; molecular biology.

## Publications

1. **J. Di Franco**, F. Krautgasser, C. Mazzella, F. Giavazzi, G. Scita, and R. Cerbino. “Structure-selected relaxation distinguishes routes to collective arrest in cell monolayers.”  
Manuscript in preparation
2. J. Kaivola, K. Punovuori, M. R. Chastney, H. Abdo, G. Follain, M. Mathieu, O. Joshi, Y. A. Miroshnikova, F. Krautgasser, **J. Di Franco**, J. R. W. Conway, S. Held, F. Bertillot, J. Hagström, A. Mäkitie, H. Irjala, S. Ventelä, H. Hamidi, G. Scita, R. Cerbino, S. A. Wickström, and J. Ivaska. “Restoring the tumour mechanophenotype of vocal fold cancer reverts its malignant properties.” *Nature Materials* **25**, 868–882 (2026).  
doi: 10.1038/s41563-025-02473-7
3. H. Abdo, L. Barzaghi, Y. Shen, E. Bellini, E. Martini, S. Magni, S. Barozzi, F. Orsenigo, D. Parazzoli, G. V. Beznoussenko, **J. Di Franco**, F. Krautgasser, J. Kaivola, M. Cinquanta, A. Lazzarin, S. Sigismund, J. Ivaska, R. Cerbino, and G. Scita. “De Novo Gene Transcription of Connexin Mediates Cytoplasmic Fluid Exchange and Flocking Transitions in Physiological and Cancerous Epithelial Systems.” *Advanced Science* **13**(6), e08648 (2026).  
doi: 10.1002/advs.202508648
4. J. M. Antúnez Domínguez, L. Pérez García, N. Rivera-Yoshida, **J. Di Franco**, D. Steiner, A. V. Arzola, M. Benítez, C. Hamngren Blomqvist, R. Cerbino, C. Beck Adiels, and G. Volpe. “*Myxococcus xanthus* for active matter studies: a tutorial for its growth and potential applications.” *Soft Matter* **21**(45), 8602–8623 (2025).  
doi: 10.1039/D5SM00284B
5. R. Ferraro, **J. Di Franco**, S. Caserta, and S. Guido. “The morphology of cell spheroids in simple shear flow.” *Frontiers in Physics* **12**, 1347934 (2024).  
doi: 10.3389/fphy.2024.1347934

## Conference Contributions

### Contributed Talk

- “Non-Universality of Jamming in Cellular Monolayers.” 9th International Soft Matter Conference, Crete, Greece (2025).

### Poster Presentations

- Statistical Physics of Living Systems, CECAM–EPFL, Lausanne, Switzerland (2025).
- ÖPG–SPS Meeting, Vienna, Austria (2025).
- Vienna Soft Matter Day, TU Wien, Vienna, Austria (2024).
- Non-equilibrium Processes in Physics and Biology, Erwin Schrödinger International Institute for Mathematics and Physics, Vienna, Austria (2024).
- Physics of Cell Fate Decisions, Institute of Science and Technology Austria, Klosterneuburg, Austria (2024).

## Teaching Experience

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Led exercise and problem-solving sessions, supported students, graded work, and assisted with examination supervision.

- **Teaching Assistant, Theoretical Physics I** — Summer semesters 2026 and 2025.
- **Teaching Assistant, Linear Algebra for Physicists (Problem Sessions)** — Winter semesters 2025 and 2024.
- **Teaching Assistant, Analysis for Physicists II** — Summer semester 2024.

## Student Supervision

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Supervised students in experimental planning, laboratory work, quantitative analysis, scientific interpretation, presentations, and thesis preparation.

- Two MSc research internships on *Myxococcus xanthus* as an active-matter system.
- One BSc research internship on *Myxococcus xanthus*.
- One BSc thesis on *Myxococcus xanthus*.
- One MSc thesis on particle tracking and microrheology.

## Research Methods and Technical Expertise

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|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Experimental systems                | Mammalian cell lines: MCF10A, HaCaT, NIH-3T3, A431, and MDCK; <i>Myxococcus xanthus</i> .                                                                                |
| Cell culture and sample preparation | Mammalian cell culture in 2D monolayers and 3D spheroids; antibody treatments; sample preparation for live-cell imaging; bacterial culture on solid and in liquid media. |
| Microscopy                          | Fluorescence, phase-contrast, and confocal microscopy; automated multi-position live-cell imaging over 24–72 hours; autofocus and environmental control.                 |
| Quantitative methods                | Microrheology; particle tracking; particle image velocimetry; differential dynamic microscopy; cell and nuclear segmentation; morphology and time-series analysis.       |
| Programming and analysis            | Python (NumPy, pandas, SciPy, scikit-learn, Matplotlib, and Seaborn); Cellpose; StarDist; Trackpy; OpenPIV; fastDDM; Fiji/ImageJ.                                        |

## Grants

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- Travel grant for Statistical Physics of Living Systems, CECAM–EPFL, Lausanne, Switzerland (2025).
- VDSP Mobility fellowship for ÖPG–SPS Meeting, Vienna, Austria (2025)

## Other responsibilities

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- Set up cell-culture laboratory, including equipment selection, procurement, installation, testing, and workflow development.
- Supported day-to-day laboratory operation through inventory and consumables management, maintenance coordination, and documentation.
- Trained new laboratory users in mammalian cell culture, including 2D monolayers and 3D spheroids, and in the culture of *Myxococcus xanthus*.
- Coordinated interdisciplinary and multi-institutional work through study planning, sample and dataset exchange, analysis, interpretation, presentations, and publication preparation.
- Served as the department poison handler.

## Certifications

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- First-aid certification, Austrian Red Cross (2024–2028).
- Poison-handling certification, University of Vienna (2025).

## Languages

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Italian (native); English (full professional proficiency); Spanish (conversational); German (basic).